

SCIENCE PROGRAM 2013

STRAND: Energy and Change TOPIC: Motion

YEAR: 10 Extension

TERM: 4 (8 weeks)

TIME (WEEK)	STUDENT OUTCOMES	PRACTICALS	TEXT REFERENCE	ASSESSMENT/HOMEWORK
1-2	Speed and Velocity Understand the difference between distance and displacement. Understand the difference between a scalar and vector quantity. List common examples of scalar and vector quantities. Be able to add vectors in one or two dimensions, using Pythagorus' Theory and the Cosine Rule. Understand the difference between speed and velocity.		p. 27 p. 28-29	ITPC Set 3-5, 19-20 ITPC Set 6
3-5	Equations of Motion Be able to use the equations of motion to make calculations to describe motion. Understand that the acceleration due to gravity near the Earth's surface is 9.80 ms^{-2} . Be able to measure motion using a ticker timer. Be able to draw graphs to represent motion - s vs t, v vs t, a vs t. Be able to calculate the gradient of a graph and understand what it represents.	Prac 2 p. 44	p. 39-40 p. 29-30, 41	ITPC Set 8-13 ITPC Set 14 Act 2.1, 2.2
6-8	Newton's Laws of Motion First Law - understand the concept of inertia. Second Law - understand that an unbalanced force will cause an object to accelerate. Be able to calculate the nett force acting in a situation, taking into account friction. Understand the difference between mass and weight. Third Law - understand the concept of action and reaction.	Prac 1 p.37 Prac 2 p. 49	p. 31-36 p. 38-40 p. 45-47	ITPC Set 15-16 Act 2.3, 2.4 Assignment - Motion Test - Motion

Texts Science Aspects Book 4
 Introduction to Physics Calculations